

Xiangyan Liu

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Education

National University of Singapore <i>Ph.D.</i> in Computer Science, School of Computing Advised by Michael Qizhe Shieh	Aug. 2024 - Present Singapore
National University of Singapore <i>M.Sc.</i> in Statistics, Department of Statistics and Data Science	Aug. 2022 - Jan. 2024 Singapore
Shanghai University of Finance and Economics <i>B.Sc.</i> in Data Science, School of Statistics and Management	Sep. 2018 - Jun. 2022 Shanghai, China

Research Interests

My research focuses on agentic and multimodal foundation models and systems, with an interest in agents for coding, computer use, and creative and professional work. I prefer solutions that are simple, generalizable, and scalable.

Experience

† National University of Singapore <i>Ph.D. student</i> , advised by Michael Qizhe Shieh	Singapore Aug. 2024 – Present
◦ Led a research project on improving VLM reasoning by developing a data augmentation technique that mixes clean and noisy trajectories during RL training to enhance reasoning robustness—with no extra cost. ◦ Co-led a research project stress-testing language models' MCP use in complex contexts, with strict state tracking (initialization, isolation, reset) and automated verification scripts to ensure safe, reliable evaluation. ◦ Contributed to a research project proposing a scalable data synthesis pipeline that improves VLM reasoning by generating challenging, verifiable data for RLVR training, yielding strong gains on hard benchmarks.	
Tencent HY <i>Research Intern</i> , working with Tianyu Pang	Singapore Mar. 2026 – Present
◦ Co-led HyCreator, an agent harness for long video generation, enabling autonomous 10-minute-scale film production with real-time interactive editing. ◦ Contributed to UniRL, an RL post-training framework for autoregressive, diffusion, and unified multimodal models, with modular rollout engines and distributed training.	
Sea AI Lab <i>Associate member</i> , working with Tianyu Pang , Chao Du , and Longxu Dou	Singapore Oct. 2024 – Feb. 2026
◦ Led a research project on State Transition Pretraining (STP), using joint forward and inverse dynamics modeling to establish visual state transitions as a new scaling axis for GUI agents. ◦ Contributed to a research project on Next-Event Prediction, a self-supervised task that improves video LLMs' temporal reasoning by predicting future events from past video segments. ◦ Contributed to a research project on feedback-conditional policy (FCP), which reframes feedback-driven learning as conditional generation, enabling LLMs to directly learn from verbal feedback.	
Moonshot AI <i>Research Intern</i> , working with Yiping Bao and Jinguo Zhu	Singapore Sep. 2025 – Feb. 2026
◦ Contributed to the OK Computer project, improving website development through iterative self-visual critics. ◦ Contributed to the development of an automatic multi-agent pipeline for agentic trajectory data generation and participated in the mid-training and post-training of Kimi-K2.5.	
Alibaba Group, Tongyi Lab <i>Research intern</i> , working with Wenmeng Zhou	Hangzhou, China Mar. 2024 – Aug. 2024
◦ Led a research project on building a code agent system that uses graph databases as interfaces to enhance LLM interaction with code repositories, enabling precise code retrieval and navigation	

Selected Work (* denotes equal contribution, † denotes student lead)

Scaling GUI Agents with Visual State Transitions

Xiangyan Liu, Kaixin Li, Haonan Wang, Biao Wu, Meng Fang, Longxu Dou, Chao Du,
Michael Qizhe Shieh, Tianyu Pang
arXiv preprint, 2026

MCPMark: A Benchmark for Stress-Testing Realistic and Comprehensive MCP Use

Zijian Wu^{†*}, Xiangyan Liu^{†*}, Xinyuan Zhang*, Lingjun Chen*, Fanqing Meng*, Lingxiao Du*, Yiran Zhao*,
Fanshi Zhang*, Yaoqi Ye, Jiawei Wang, Yufan Yang, Jinjie Ni, Zirui Wang, Arvin Xu, Michael Qizhe Shieh
International Conference on Learning Representations (ICLR), 2026

NoisyRollout: Reinforcing Visual Reasoning with Data Augmentation

Xiangyan Liu*, Jinjie Ni*, Zijian Wu*, Chao Du, Longxu Dou, Haonan Wang, Tianyu Pang, Michael Qizhe Shieh
Advances in Neural Information Processing Systems (NeurIPS), 2025

CodexGraph: Bridging Large Language Models and Code Repositories via Code Graph Databases

Xiangyan Liu*, Bo Lan*, Zhiyuan Hu, Yang Liu, Zhicheng Zhang, Fei Wang, Michael Shieh, Wenmeng Zhou
North American Chapter of the Association for Computational Linguistics (NAACL), 2025

Towards Robust Multi-Modal Reasoning via Model Selection

Xiangyan Liu*, Rongxue Li*, Wei Ji, Tao Lin
International Conference on Learning Representations (ICLR), 2024

Online Distillation-Enhanced Multi-Modal Transformer for Sequential Recommendation

Wei Ji*, Xiangyan Liu*, An Zhang, Yinwei Weo, Yongxin Ni, Xiang Wang
ACM International Conference on Multimedia (ACM MM), Oral, 2023

Service

Conference Reviewer: ICML (2026), ICLR (2025, 2026, 2027), NeurIPS (2024, 2025), ACM MM (2023, 2024)

Skills

Programming Languages: Python, R, SQL, Shell Script, Scala, LaTeX

Frameworks and Tools: PyTorch (Distributed Training), SGLang, vLLM, verl, slime, Ray, Docker

Languages: Mandarin (native), English (fluent)

Teaching

Teaching Assistant

CS5425/CS4225 - Big Data Systems for Data Science
National University of Singapore

Singapore
Autumn 2025

Teaching Assistant

CS5260 - Neural Networks and Deep Learning II
National University of Singapore

Singapore
Spring 2025